The Reuleaux Platonic Solids

Balls as wide as the solid on every corner: four ball polyhedra beside the Reuleaux tetrahedron, named by their look, with their fewest face colours and their widths; dual pairs have exactly equal width ranges

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Version 1 · 21 October 2026 · DICTO DOI: reserved on publication · Licence CC BY-SA 4.0 · Scripts, checks and figure: https://github.com/DICTOR-Master/kaleidohedra/tree/0294763/studies/2026-10-21-reuleaux-platonic (research branch, commit 0294763; also attached to this record)
The construction (a ball on every corner of a Platonic solid, radius = the solid's diameter) and the names by look (Reuleaux octahedron, cube, icosahedron and dodecahedron) are DICTO's idea; computation, proofs and drafting with Claude (Anthropic).

Abstract. DICTO's idea: put a ball on every corner of a Platonic solid, each as wide as the solid (radius = its diameter), and keep what all the balls share. On the tetrahedron this is the classical Reuleaux tetrahedron, and our numerical volume agrees with the known closed form to 1.1×10^{-8}. On the other four it gives ball polyhedra which DICTO names by their look: balls on the cube's corners make the **Reuleaux octahedron**, on the octahedron's the **Reuleaux cube**, on the dodecahedron's the **Reuleaux icosahedron** and on the icosahedron's the **Reuleaux dodecahedron**. Each ball gives one face, and two faces meet along an arc exactly when their balls' centres are joined by an edge, so each body has the face pattern of the dual solid; the fewest colours with no two neighbours alike are 4, 2, 3, 3 and 4. They are not Reuleaux polyhedra in Sallee's sense: the corners are not vertices of the body, and the diametric pairs are fewer than $2m - 2$. We prove that every width is at least the diameter, with equality through two opposite corners, and that along the axis through a face centre the width is exactly $2\sqrt{(3/4 + r^2)} - 2r$ at circumradius ½ (r the inradius). Dual solids have the same inradius-to-circumradius ratio, so dual pairs share this width exactly: $(\sqrt{30} - \sqrt{3})/3 \approx 1.248392$ for the Reuleaux octahedron and cube, $(\sqrt{(50 + 2\sqrt{5})} - \sqrt{(5 + 2\sqrt{5})})/\sqrt{15} \approx 1.110989$ for the Reuleaux icosahedron and dodecahedron. That the face axis gives the largest width is found numerically and not proved.

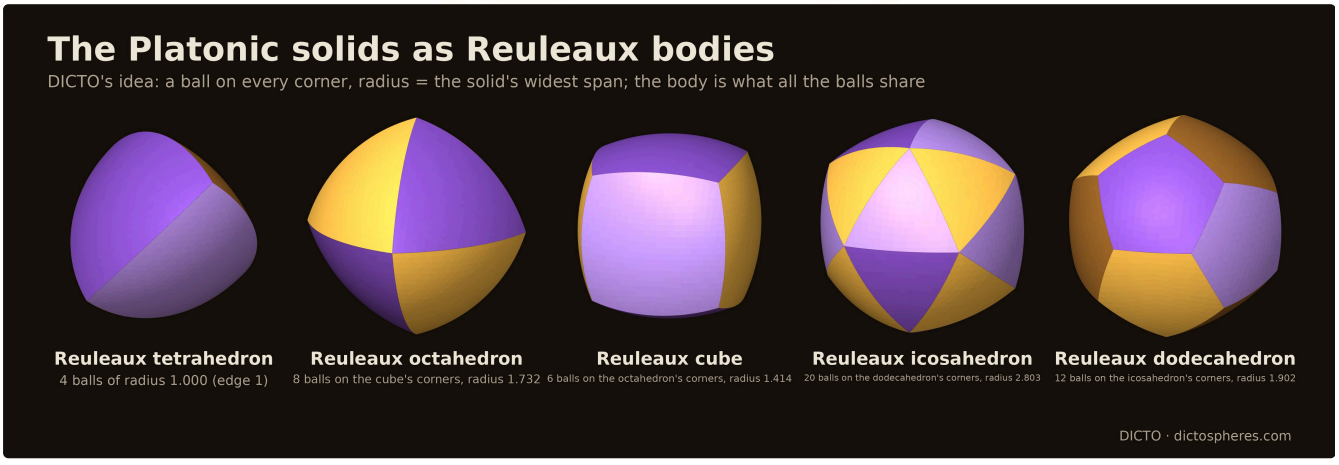


Figure 1. The five bodies at solid edge 1, each with its faces coloured with the fewest colours that keep neighbours apart (gold, purple, lilac, bronze). Under each: the number of balls and their radius, which is the solid's diameter. Meshes from reuleaux_platonic.py.

1. The construction

For a Platonic solid S with corner set X and diameter D (its widest span), DICTO's body is B = $\cap_{v \in X}$ B(v, D), the common part of the closed balls of radius D centred on the corners. In the language of [1, 2] it is a ball polyhedron. For the tetrahedron D is the edge and B is the Reuleaux tetrahedron [2, 3]. For the cube, octahedron, dodecahedron and icosahedron D is twice the circumradius R: the widest spans join opposite corners. Below, all lengths are at D = 1.

Every corner lies in B (no two corners are more than D apart), and the surface of B is made of one spherical face from each ball [3]: the part of the sphere of radius D about v that bounds B, lying across the body from v. DICTO named the bodies by their look, and the look is explained by the faces. The checks sample, for every pair of balls, the circle where their spheres meet, and find that two faces meet along an arc exactly when their centres are joined by an edge of S (Table 1). So the faces of B correspond to the corners of S and meet as those corners are joined, which is how the faces of the dual solid meet: the balls on the cube's 8 corners give 8 faces meeting like the octahedron's, and so on.

Body (DICTO's name)	Balls on	Faces	Arcs	Diametric pairs (2m − 2)	Fewest colours	Width range	Volume ≈
Reuleaux tetrahedron	tetrahedron	4	6	6 (6)	4	1 .. 1.024944	0.42216
Reuleaux octahedron	cube	8	12	4 (14)	2	1 .. 1.248392	0.64940
Reuleaux cube	octahedron	6	12	3 (10)	3	1 .. 1.248392	0.68764
Reuleaux icosahedron	dodecahedron	20	30	10 (38)	3	1 .. 1.110989	0.57188
Reuleaux dodecahedron	icosahedron	12	30	6 (22)	4	1 .. 1.110989	0.59679

Table 1. The five bodies at diameter 1. Faces: one per ball; arcs: pairs of faces meeting along an arc (numerical; equal to the solid's edges); diametric pairs: corner pairs at distance D, exact, with the bound 2m − 2 for m corners in brackets; fewest colours with no two neighbouring faces alike; width range: the minimum is proved (§3), the maximum is the largest width found numerically, proved to be attained (§3) but not proved to be the largest; volume by radial quadrature (numerical).

2. The tetrahedron check, and what is not a Reuleaux polyhedron

On the tetrahedron the construction is the classical Reuleaux tetrahedron [2, 3]. Its volume at width 1 is $8\pi/3 + \sqrt{2}/4 - (27/4)\arccos(1/3)$ (Harbourne's formula, as quoted by Hynd [3]); our quadrature gives 0.4221577, equal to it to 1.1×10^{-8}. Its widths run from 1 to 1.024944; the largest found is attained along the axis through the midpoints of two opposite edges, where the same argument as in Theorem 1 below gives exactly $\sqrt{3} - \sqrt{2}/2$: the opposite edges are $1/\sqrt{2}$ apart and every corner is ½ from that axis.

Hynd [3] and Montejano and Roldán-Pensado [2] call B(X) a Reuleaux polyhedron when X has the largest possible number of diametric pairs, 2m − 2 for m points (the Vázsonyi bound, proved by Grünbaum, Heppes and Straszewicz [3]), or, after Sallee, when the vertices of B are exactly the points of X [2]. The tetrahedron has $6 = 2 \times 4 - 2$. The other four do not qualify (exact checks): their diametric pairs are only the opposite corners, 4, 3, 10 and 6, against 14, 10, 38 and 22; and each corner lies on the sphere of exactly one other ball (the opposite corner's), so it is not a vertex of B, where at least three spheres meet [2]. They are ball

polyhedra, not Reuleaux polyhedra, and the smoothing of edge pairs that turns a Reuleaux polyhedron into a body of constant width [2, 3] does not apply to them as it stands.

3. Widths

The width of B in a direction u is the distance between the two planes at right angles to u that touch B.

> **Proposition 1 (the smallest width is the diameter).** For every one of the five bodies every width is at least 1, and the width along the line through any two corners at distance 1 is exactly 1.

Proof. By Pál's theorem, as cited in [2], the corner set X (of diameter 1) lies in a body K of constant width 1. K has diameter 1 and contains X, so every point of K is within 1 of every corner: $K \subset B$, and every width of B is at least 1. Now let v, w be corners with $|v - w| = 1$ and $u = v - w$. B lies in the ball of radius 1 about w, so $u \cdot x \le u \cdot w + 1 = u \cdot v$ on B, and v lies in B; so the largest value of $u \cdot x$ on B is $u \cdot v$, and in the same way the smallest is $u \cdot w$. The width is $u \cdot v - u \cdot w = 1$. ∎

> **Theorem 1 (the width along a face axis).** Let S be a cube, octahedron, dodecahedron or icosahedron with circumradius ½ (diameter 1) and inradius r. Along the axis through its centre and a face centre the width of B is exactly $2\sqrt{(3/4 + r^2)} - 2r$.

Proof. Let u be the unit vector to the face centre and $h_v = u \cdot v$ the height of a corner v. The rotations of S about the axis (3-, 4- or 5-fold) map B to itself and fix u, so they map the set of points of B where $u \cdot x$ is largest to itself; that set is convex, and its centroid lies on the axis. So the largest value of $u \cdot x$ on B is reached at a point zu of the axis. Since $|v|^2 = ¼$, $|zu - v|^2 = ¼ + z^2 - 2z\,h_v$, and for $z > 0$ this is largest where h_v is smallest. The plane of the opposite face touches S, so every corner has $h_v \ge -r$, with equality exactly at the corners of that face (checked exactly). Hence zu lies in B exactly when $¼ + z^2 + 2zr \le 1$, that is $z \le \sqrt{(3/4 + r^2)} - r$. S and B are centrally symmetric, so the width is twice this. ∎

Dual pairs. The width depends only on r, and $r = ½ \times (r/R)$. Dual solids have the same ratio r/R; the exact checks find $(r/R)^2 = 1/3$ for the cube and the octahedron and $(5 + 2\sqrt{5})/15$ for the dodecahedron and the icosahedron. So the Reuleaux octahedron and the Reuleaux cube have exactly the same face-axis width, and so do the Reuleaux icosahedron and the Reuleaux dodecahedron:

- cube, octahedron: $r^2 = 1/12$, $2\sqrt{(3/4 + r^2)} = \sqrt{30}/3$ and $2r = \sqrt{3}/3$, so **$w = (\sqrt{30} - \sqrt{3})/3 \approx 1.248392$**;
- dodecahedron, icosahedron: $r^2 = (5 + 2\sqrt{5})/60$, $(2\sqrt{(3/4 + r^2)})^2 = (50 + 2\sqrt{5})/15$ and $(2r)^2 = (5 + 2\sqrt{5})/15$, so **$w = (\sqrt{(50 + 2\sqrt{5})} - \sqrt{(5 + 2\sqrt{5})})/\sqrt{15} \approx 1.110989$**.

The largest width (numerical). The width was computed without a mesh, from the exact candidate points of the support function (ball caps, the circles where two spheres meet, the points where three meet), over 3000 random directions with local refinement (widths.py). The largest widths found are 1.248391589161 for both the Reuleaux octahedron and cube and 1.110988655456 for both the Reuleaux icosahedron and dodecahedron, equal to the face-axis values of Theorem 1 to 12 digits, and $1.024944026382 = \sqrt{3} - \sqrt{2}/2$ for the tetrahedron. That

the face axis (or, for the tetrahedron, the edge-midpoint axis) gives the global maximum is therefore numerical; Theorem 1 proves only the value along that axis. Since the smallest width is 1 (Proposition 1), the width ranges of each dual pair, and so their spreads (largest/smallest), are equal, given that numerical maximum.

4. Faces and colours

Because two faces meet along an arc exactly when their corners are joined by an edge of S (§1), colouring the faces so that no two neighbours share a colour is colouring the corners of S along its edges. The fewest colours, found by exhaustive search in the checks, are 4 for the Reuleaux tetrahedron (any two of its 4 faces meet), 2 for the Reuleaux octahedron (the cube's corners split into two sets with no edge inside either), 3 for the Reuleaux cube, 3 for the Reuleaux icosahedron and 4 for the Reuleaux dodecahedron, as in Figure 1. These are the numbers DICTO read from the look: 4, 2, 3, 3, 4.

5. Prior art and what is claimed

- **Known, credited:** the Reuleaux tetrahedron and its volume (Harbourne, as quoted by Hynd [3]); ball polyhedra in general (Bezdek, Lángi, Naszódi and Papez [1]); Reuleaux polyhedra after Sallee [4], Meissner's solids and the Meissner polyhedra built from Reuleaux polyhedra (Montejano and Roldán-Pensado [2]); Pál's theorem (as cited in [2]); Hynd's perimeter and volume formulas for Reuleaux polyhedra, with a family of Reuleaux pyramids on regular n-gons for odd n [3]. The construction B(X) itself is the standard one of [1, 2, 3].
- **Claimed here:** the four bodies on the cube, octahedron, dodecahedron and icosahedron (DICTO's idea and names), the fact that they are not Reuleaux polyhedra, their face patterns and fewest colours (the arcs found numerically), Proposition 1 for them, Theorem 1 and the equal face-axis widths of dual pairs with their closed forms, and the numerical width ranges and volumes.

The three library papers on these bodies [1, 2, 3] were searched in full for cube, octahedron, dodecahedron, icosahedron, Platonic and regular: the only regular solid they use is the regular tetrahedron or simplex (for the Reuleaux and Meissner tetrahedra, Hynd's pyramids and a lemma of [1]). Sallee's paper [4] is in the library as a link only and was not read. In the sources checked (the full text of the 81 library sources with a text layer listed in the appendix, searched on 21 October 2026; the library's 173 link-only entries, Sallee's among them, were not searched) we did not find the bodies on the other four Platonic solids or their widths.

Data and reproducibility

In studies/2026-10-21-reuleaux-platonic: reuleaux_checks.py (log reuleaux-checks.log, 45 PASS, no FAIL), in two parts. Part A is exact ($\mathbb{Q}(\sqrt{5})$ with Python fractions): the diameters, diametric pairs and the corners lying on one other sphere; the corner heights along a face axis; $(r/R)^2$ of each solid; the closed forms of Theorem 1; and the tetrahedron's edge-midpoint axis. Part B is numerical (float64): the face-axis widths against widths.log, the volumes by radial Gauss–Legendre quadrature (N = 1000 and 2000 agree to within 6×10^{-7}), the Reuleaux tetrahedron's volume against Harbourne's formula, and the arcs (7200 points on each circle, margin 10^{-9}) with the fewest colours by backtracking. widths.py (widths.log): the width search. reuleaux_platonic.py and reuleaux_compose.py: the meshes (manifold3d), the colouring and Figure 1.

References

1. K. Bezdek, Z. Lángi, M. Naszódi, P. Papez, Ball-polyhedra (2007), arXiv:1110.4329.
2. L. Montejano, E. Roldán-Pensado, Meissner polyhedra (2017), arXiv:1608.06354.
3. R. Hynd, The perimeter and volume of a Reuleaux polyhedron (2023), arXiv:2310.08709.
4. G. T. Sallee, Reuleaux polytopes, Mathematika 17 (1970) 315–323, doi:10.1112/S0025579300002977 (link only; not read).

Appendix. The reference library searched

The library's manifests list 267 entries, of which 94 have a local file; these give the 81 distinct sources with a text layer listed below and 13 page images. The full text of the 81 sources was searched on 21 October 2026 for: Reuleaux, Meissner, ball polyhedron, ball-polyhedra, constant width, intersection of balls, and, in the three papers that matched, cube, octahedron, dodecahedron, icosahedron, Platonic and regular. Only [1, 2, 3] matched the first terms. The 13 page images without a text layer (the patents of Fuller, Snelson, Pearce, Baer, Hogan, Bergman and Lalvani, and Fedorov 1885) and the 173 link-only entries were not searched.

Author(s)	Year	Title
Luca Pacioli, Leonardo da Vinci (ill.)	1509	Divina proportione
Albrecht Dürer	1525	Underweysung der Messung mit dem Zirckel und Richtscheyt
Wenzel Jamnitzer, Jost Amman (engr.)	1568	Perspectiva corporum regularium
Johannes Kepler	1596	Prodromus dissertationum cosmographicarum continens Mysterium cosmographicum
Johannes Kepler	1619	Harmonices mundi libri V
E. S. Fedorov (chapter 13, English translation)	1885	Elements of the Theory of Figures, chapter 13
Euclid, T. L. Heath (tr.)	1908	The Thirteen Books of Euclid's Elements, Vol. III (Books X-XIII)
M. Goldberg	1937	A class of multi-symmetric polyhedra
Haresh Lalvani	1990	Transformational part-count in layered octahedral-tetrahedral truss configurations (NASA TM-102635)
P. W. Messer	1995	Stellations of the rhombic triacontahedron and beyond
M. Baake, F. Gähler	1998	Symmetry structure of the Elser-Sloane quasicrystal
T. C. Hales	1998	An overview of the Kepler conjecture
M. Baake	1999	A guide to mathematical quasicrystals
O. Ogievetsky, Z. Papadopolos	1999	On Quasiperiodic Space Tilings, Inflation and Dehn Invariants
U. Betke, M. Henk	1999	Densest lattice packings of 3-polytopes
Z. Papadopolos, O. Ogievetsky	1999	On Inflation Rules for Mosseri-Sadoc Tilings
J. H. Conway, C. Radin, L. Sadun	1999	On angles whose squared trigonometric functions are rational
R. V. Moody	2000	Model sets: a survey
I. Hafner, T. Zitko	2002	Rhombic polyhedra: from dissection of the cube to space filling with prolate rhombohedra and rhombic dodecahedra of the second kind
M. Senechal	2004	Donald and the golden rhombohedra
Wikipedia contributors (2007 mirror)	2007	Kyushindo (2007 version)
S. Kabai	2007-2015	Wolfram Demonstrations: 54-Faced Space-Filling Polyhedron (2007), Golden Octet Truss (2007), A Rhombic Dodecahedron Made of Rhombic Triacontahedra (2008), Relating a Rhombic Triacontahedron and a Rhombic Dodecahedron (2008), Cluster of 14 Rhombic Triacontahedra (2015)
K. Bezdek, Z. Lángi, M. Naszódi, P. Papez	2007	Ball-polyhedra
M. Koca, M. Al-Ajmi, N. O. Koca	2009	Grand antiprism and quaternions
M. Koca, M. Al-Ajmi, N. O. Koca	2009	Quaternionic representation of snub 24-cell and its dual polytope derived from E8 root system
S. Torquato, Y. Jiao	2009	Dense packings of polyhedra: Platonic and Archimedean solids
B. Grünbaum	2010	The Bilinski dodecahedron, and assorted parallelohedra, zonohedra, monohedra, isozonohedra and otherhedra
P. Kramer	2010	Gateways towards quasicrystals
A. M. Cutler, E. Schulte	2011	Regular polyhedra of index two, I

D. Frettlöh	2011	Icosahedral tilings in R3: the ABCK tilings (draft chapter)
J. Koskinen	2011	Unusual tilings and transformations
J. de Graaf, R. van Roij, M. Dijkstra	2011	Dense regular packings of irregular non-convex particles
M. Koca, N. O. Koca, M. Al-Ajmi	2011	Branching of the W(H4) polytopes and their dual polytopes under the Coxeter groups W(A4) and W(H3) represented by quaternions
M. Koca, N. O. Koca, M. Al-Barwani	2011	Snub 24-cell derived from the Coxeter-Weyl group W(D4)
T. Etzion	2011	Product constructions for perfect Lee codes
A. Garber, A. Gavrilyuk, A. Magazinov	2012	The Voronoi conjecture for parallelohedra with simply connected delta-surface
J. C. Lagarias, C. Zong	2012	Mysteries in packing regular tetrahedra
M. Baake, U. Grimm	2012	Mathematical diffraction of aperiodic structures
M. Koca, N. O. Koca, R. Koc	2012	Affine Coxeter group Wa(A4), quaternions, and decagonal quasicrystals
P. F. Damasceno, M. Engel, S. C. Glotzer	2012	Crystalline assemblies and densest packings of a family of truncated tetrahedra and the role of directional entropic forces
R. Gabbrielli, Y. Jiao, S. Torquato	2012	Families of tessellations of space by elementary polyhedra via retessellations of face-centered-cubic and related tilings
S. Torquato, Y. Jiao	2012	Organizing principles for dense packings of nonspherical hard particles: not all shapes are created equal
M. Kléman	2013	Defects in quasicrystals, revisited: I-flips, approximants, phason defects
N. Fujita, H. Takano, A. Yamamoto, A.-P. Tsai	2013	Cluster packing geometry for Al-based F-type icosahedral alloys
P.-P. Dechant, J. Wardman, T. Keef, R. Twarock	2014	Viruses and fullerenes: symmetry as a common thread?
A. Campello, G. C. Jorge, J. E. Strapasson, S. I. R. Costa	2015	Perfect codes in the lp metric
K. Deguchi, M. Nakayama, S. Matsukawa, K. Imura et al.	2015	Crystal structure of superconducting 1/1 cubic Au-Ge-Yb approximant with Tsai-type cluster
M. Baake, D. Damanik, U. Grimm	2015	What is aperiodic order?
M. Firsching	2015	Computing maximal copies of polytopes contained in a polytope
N. O. Koca, M. Koca, A. Al-Mukhaini, A. Al-Qanobi	2015	Quaternionic representations of the pyritohedral group, related polyhedra and lattices
R. Twarock, M. Valiunas, E. Zappa	2015	Orbits of crystallographic embedding of non-crystallographic groups and applications to virology
A. T. Cadotte, J. Dshemuchadse, P. F. Damasceno, R. S. Newman, S. C. Glotzer	2016	Self-assembly of a space-tessellating structure in the binary system of hard tetrahedra and octahedra
C. Hann, J. E. S. Socolar, P. J. Steinhardt	2016	Local growth of icosahedral quasicrystalline tilings
R. M. K. Dietl, J.-H. Eschenburg	2016	The icosahedral quasiperiodic tiling and its self-similarity
L. Montejano, E. Roldán-Pensado	2017	Meissner polyhedra
A. Stergiou	2018	Bootstrapping hypercubic and hypertetrahedral theories in three dimensions
A. Garber	2019	Voronoi conjecture for five-dimensional parallelohedra
D. Poland, S. Rychkov, A. Vichi	2019	The conformal bootstrap: theory, numerical techniques, and applications
R. Twarock, A. Luque	2019	Structural puzzles in virology solved with an overarching icosahedral design principle
T. Ishimasa	2019	Icosahedral quasicrystal, 1/1 and 2/1 approximants in Zn-based ternary alloys containing Au and Yb/Tb
A. Al-Siyabi, N. O. Koca, M. Koca	2020	Icosahedral Polyhedra from D6 lattice and Danzer's ABCK tiling
M. Koca, R. Koc, N. O. Koca, A. Al-Siyabi	2020	Icosahedral tiling with dodecahedral structures
N. O. Koca, R. Koc, M. Koca, A. Al-Siyabi	2020	Dodecahedral Structures with Mosseri-Sadoc Tiles
V. R. Krishnamurthy, E. Akleman, S. G. Subramanian et al.	2020	Bi-axial woven tiles: interlocking space-filling shapes based on symmetries of bi-axial weaving patterns
V. R. Krishnamurthy, E. Akleman, S. G. Subramanian, M. Ebert, J. Cui, C. Fu, C. Starrett	2022	Geometrically interlocking space-filling tiling based on fabric weaves
B.-X. Lao, S. Rychkov	2023	3D Ising CFT and exact diagonalization on icosahedron: the power of conformal perturbation theory

D. Smith, J. S. Myers, C. S. Kaplan, C. Goodman-Strauss	2023	An aperiodic monotile
R. Hynd	2023	The perimeter and volume of a Reuleaux polyhedron
R. Akpanya et al.	2024	Topologically interlocking blocks inside the tetroctahedrille
I. J. Dejter, L. R. Fuentes, C. A. Araujo	2025	Perfect truncated-metric codes in lattice graphs and superlattice graphs
S. Rychkov	2025	Conformal bootstrap: from Polyakov to our times
M. Kovačević	2026	An isodiametric theorem and lattice diameter-perfect codes in A3
R. Al Raisi, N. O. Koca, M. Koca, R. Koc	2026	Modified Mosseri-Sadoc tiles from D6
Wikipedia contributors	2026	Kyūshindō
Wikipedia contributors	2026	Kenshiro Abbe
Wikipedia contributors	2026	Mitsusuke Harada
Wikipedia contributors	2026	n-flake (Sierpinski icosahedron)
Wikipedia contributors	2026	Fullerene
Wikipedia contributors	2026	Goldberg polyhedron
Wikipedia contributors	2026	Tetrahedral-octahedral honeycomb (incl. the gyrated form)
Wikipedia contributors	2026	Crystal twinning (incl. the spinel law)